

FSAC results

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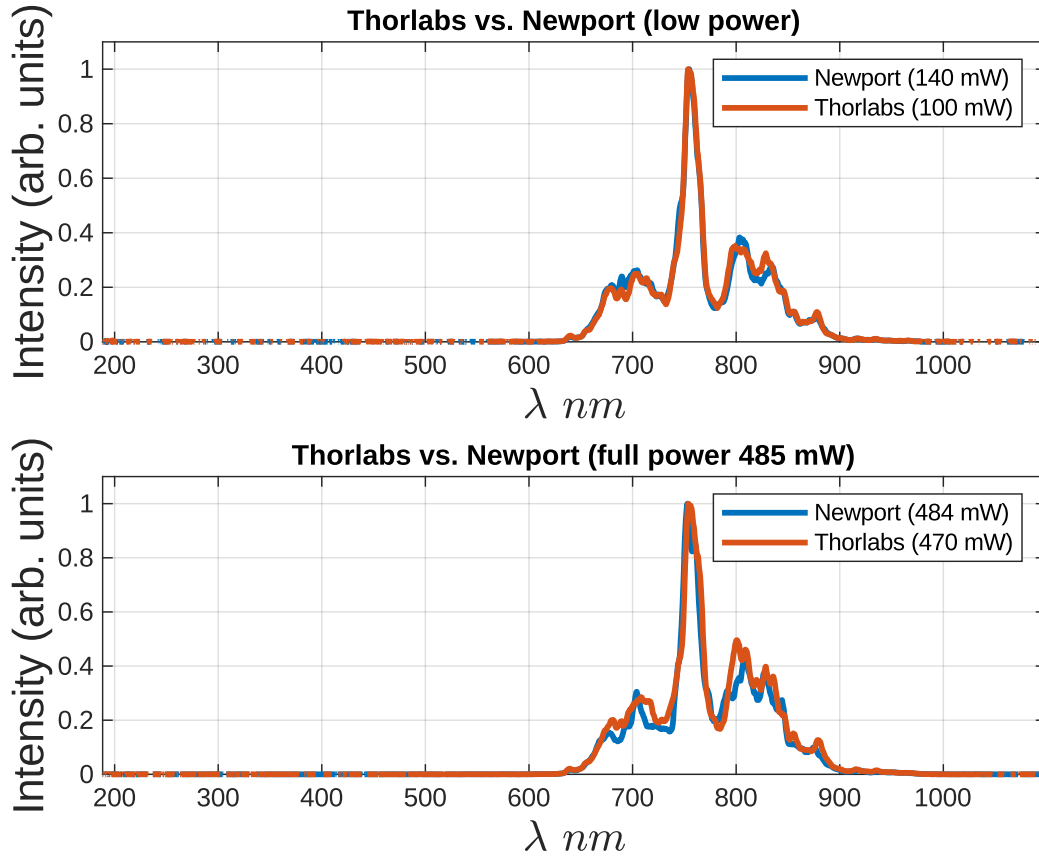
June 11, 2024

1 So many different mirror sets

All of you are aware of the large number of different mirrors that me and Asad have been using with the 10 fs Octavius laser. Long story short, it comes down to two different sets of mirrors:

- Newport 10B20EAG.1 fs optimized silver mirrors.
- Thorlabs PF10-03-P01 protected silver mirrors.

The first task was to compare the Spectra obtained after only one bounce off of the mirror to see if there clipping or a noticeable difference between each other and the previously measured spectrum. There is no



apparent clipping nor differences between the spectra nor a noticeable difference with the spectrum we have always had, but I noticed that the full power is very low compared to what it was before ($\sim 650mW$).

2 Autocorrelation trace as a function of chirped mirror bounces

The current layout using Thorlabs' UMC05-15FS chirped mirrors and the Femto-Second-Auto-Correlator (FSAC) can be seen in figure 1. According to the FSAC manual, a clear indication of a chirped pulse hitting the FSAC's non-linear detector is that the trace will start off-setting at the central peak, followed by the lateral wings of the trace (see figure 13, page 17). At this distance, it's clear that the number of bounces on the chirped mirrors that minimizes chirp is 8 (see figs. 2 to 7)

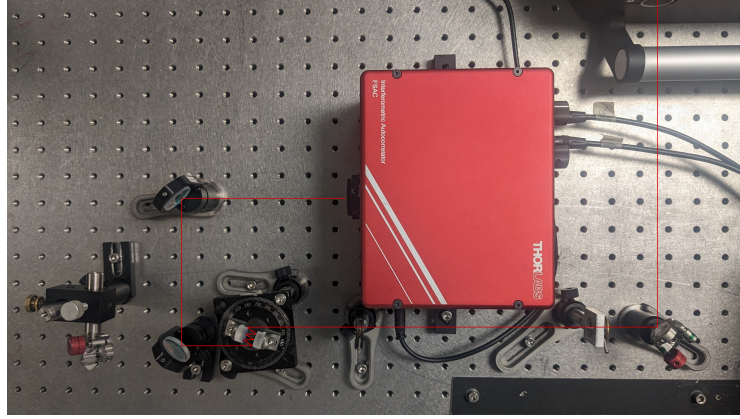


Figure 1: Layout of the pulse duration experiment. Notice that this is an old picture with the incorrect dielectric mirrors Thorlabs recommended a while ago, but the layout is the same (albeit the slightly different positions of the mirrors after the chirped mirror stage). The new layout also includes a half waveplate to ensure that the beam is P-polarized before it enters the FSAC.

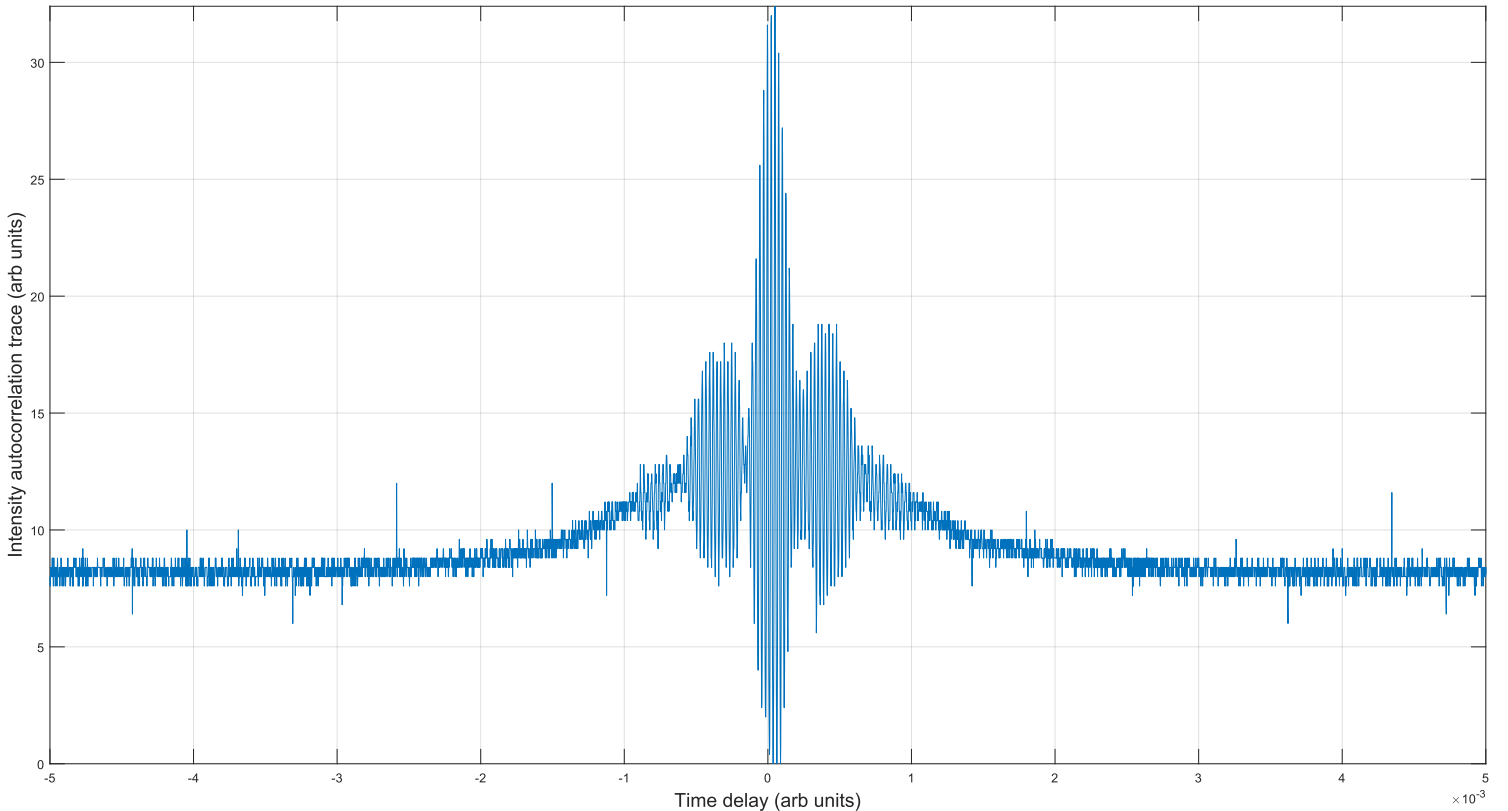


Figure 2: Interferometric trace for $n = 2$ bounces.

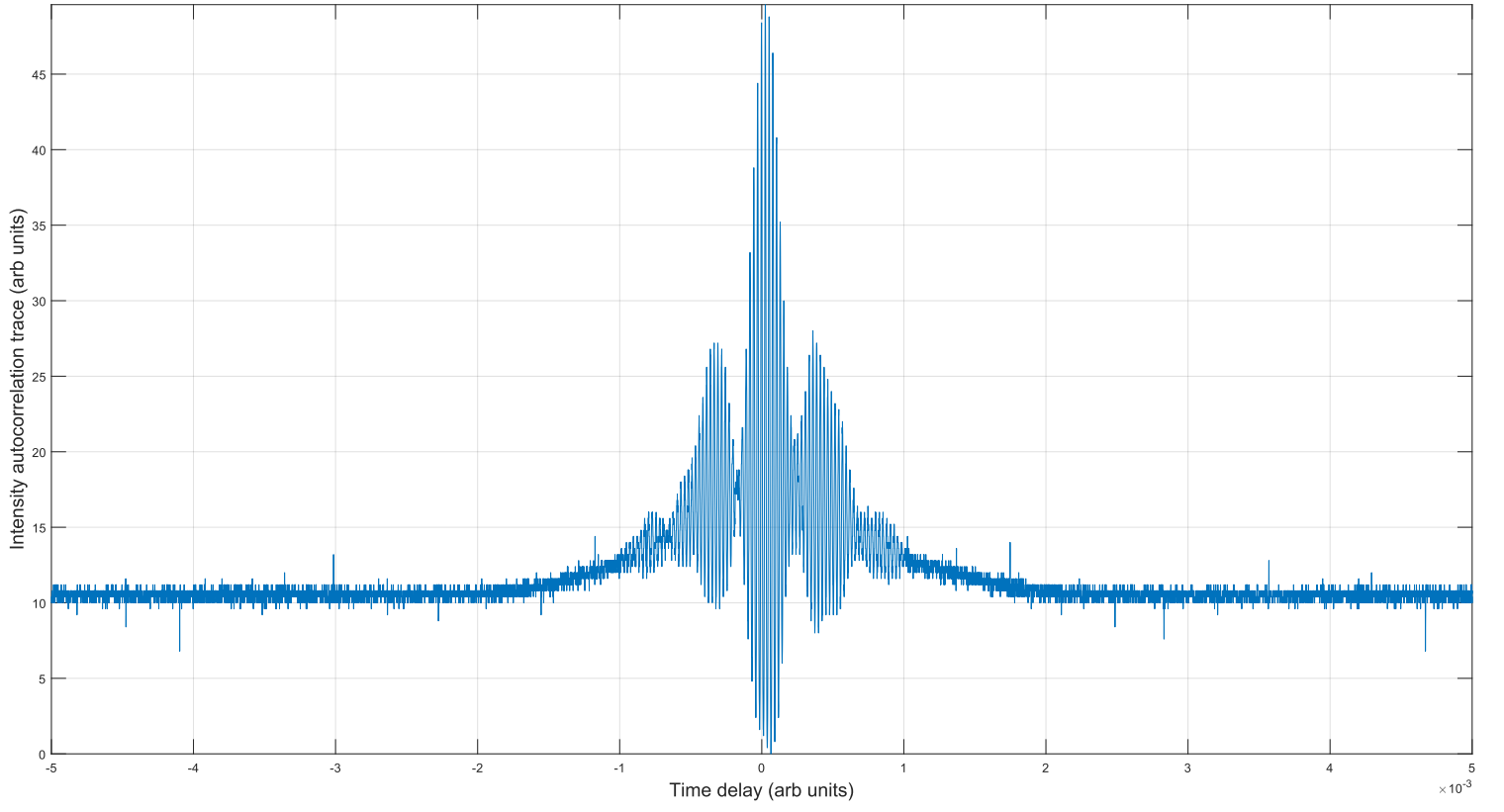


Figure 3: Interferometric trace for $n = 4$ bounces.

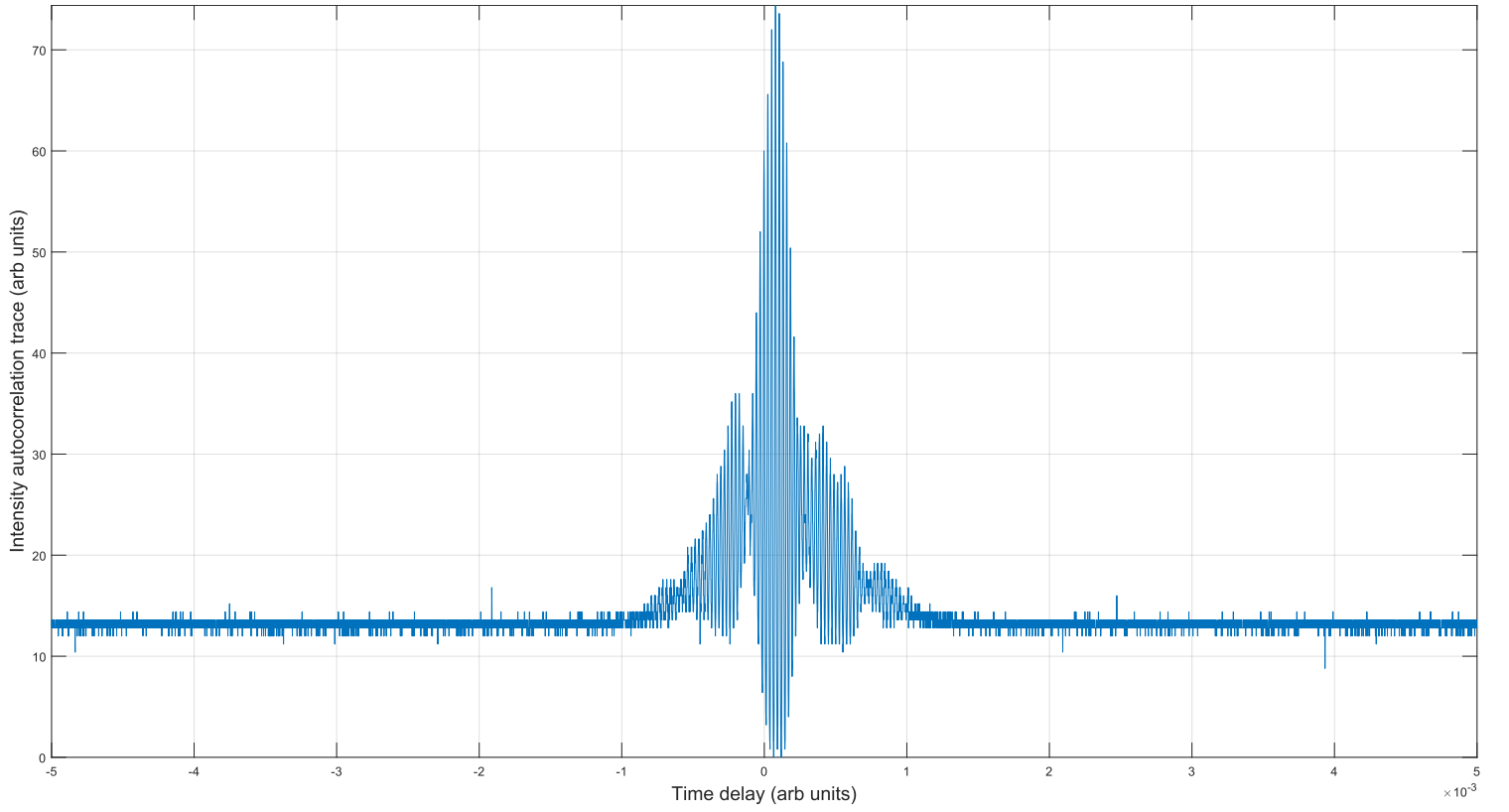


Figure 4: Interferometric trace for $n = 6$ bounces.

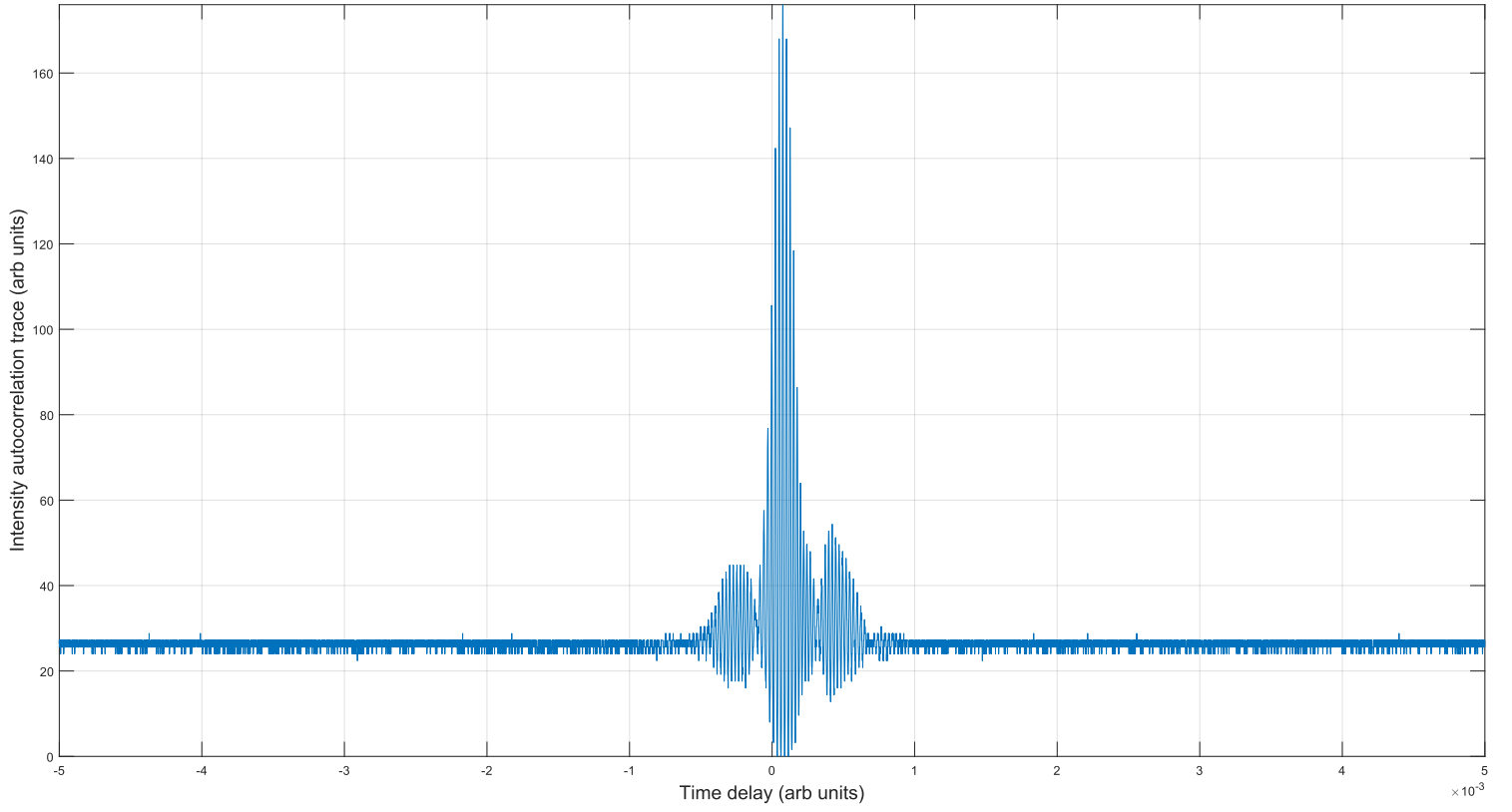


Figure 5: Interferometric trace for $n = 8$ bounces. Note that this looks very similar to the trace that Thorlabs reports in the manual (figure 7, page 10), except for the lateral wings.

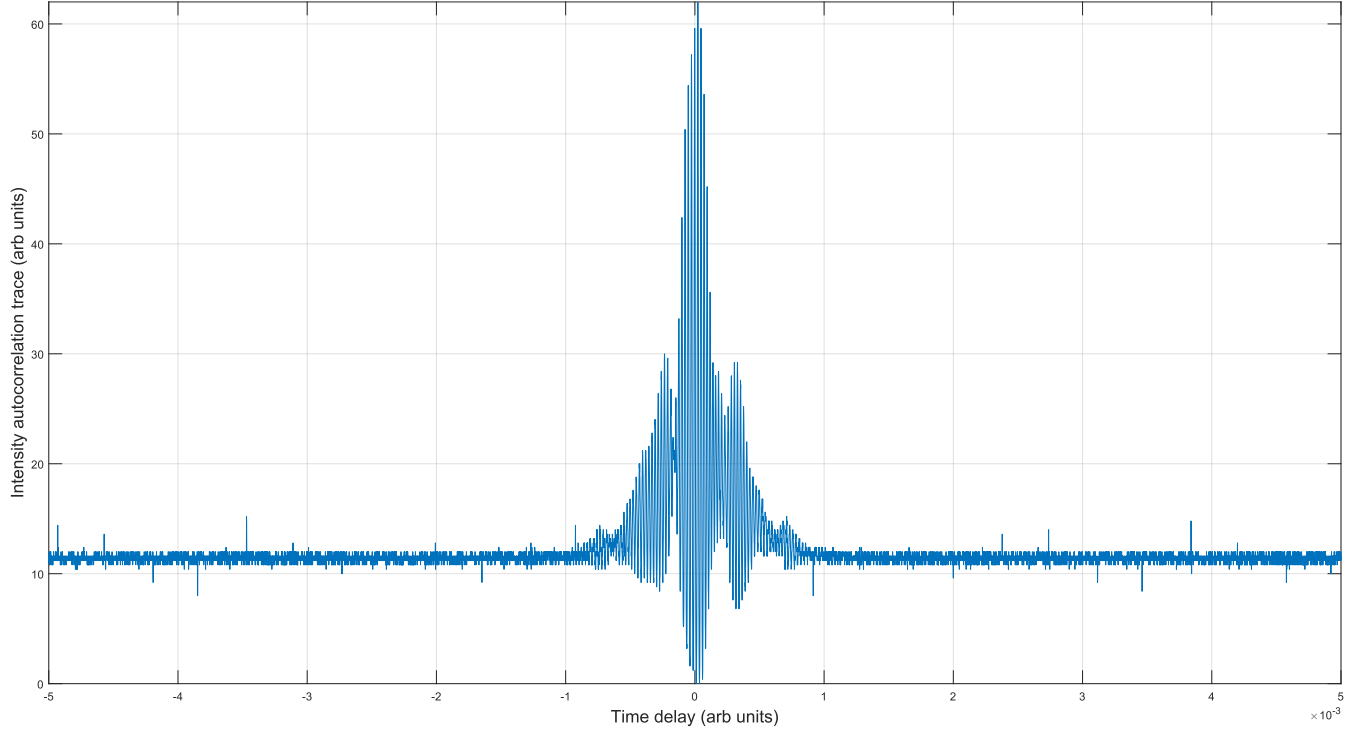


Figure 6: Interferometric trace for $n = 10$ bounces.

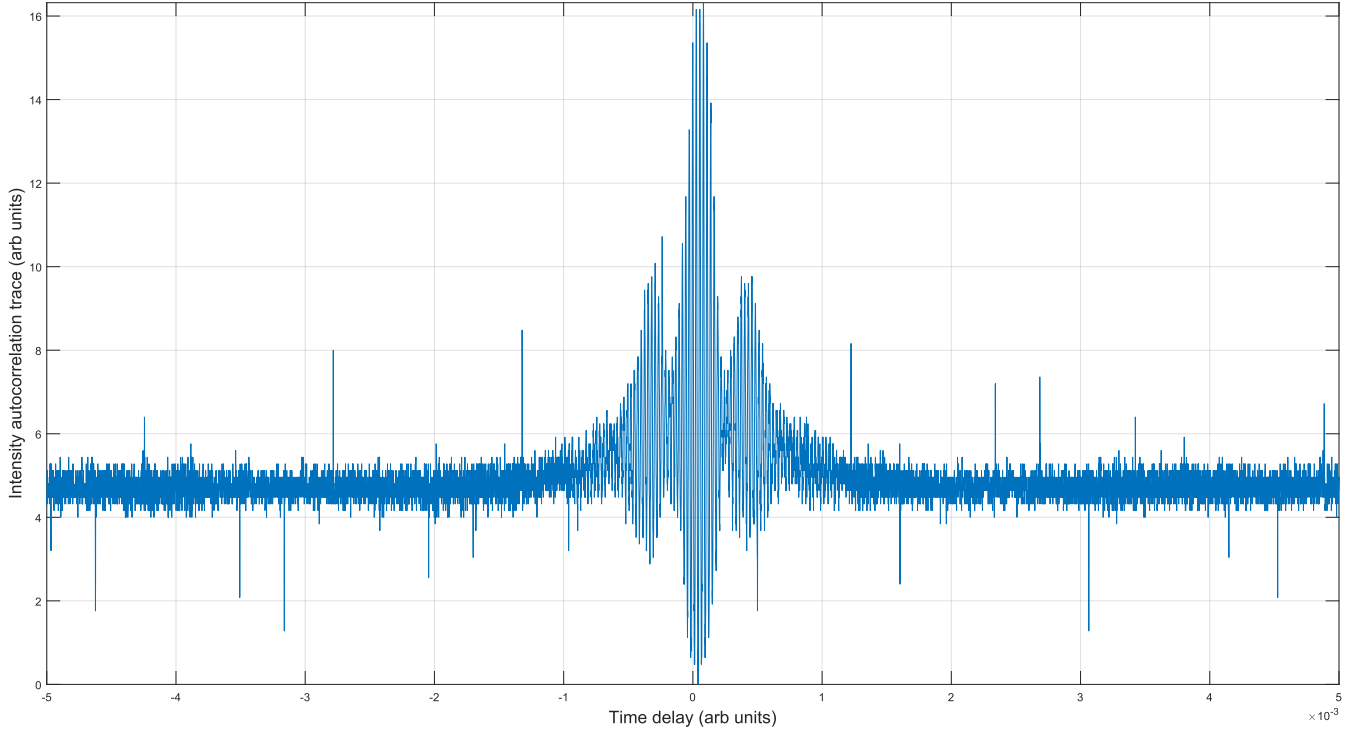


Figure 7: Interferometric trace for $n = 12$ bounces. Note that the peak-to-offset ratio goes down as chirp increases.

3 Pulse duration

With these traces, we can agree that the chirp is minimized for $n = 8$ bounces off the chirped mirrors. This also happens for the same number of chirped mirror bounces when using Thorlabs' P01 mirrors (see figure 8) Using these two traces, Thorlabs' little snippet gives the pulse durations

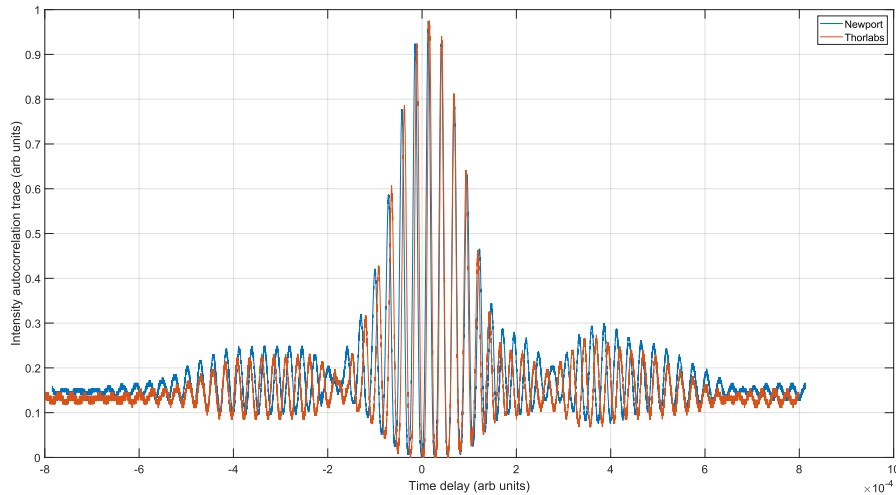


Figure 8: Interferometric trace for $n = 8$ chirped mirror bounces for both Newport and Thorlabs' silver mirrors.

Newport	Gaussian pulse FWHM : 12.4113 fs , sech^2 pulse FWHM : 11.3754 fs	(1a)
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Thorlabs	Gaussian pulse FWHM : 12.7195 fs , sech^2 pulse FWHM : 11.6579 fs	(1b)
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4 Glass window

Following Herman's suggestion, Asad measured the GDD added to the pulse when it goes through an observation window on a 3 in flange. The value turned out to be $\text{GDD}_{\text{window}} = (121 \pm 6) \text{ fs}^2$ for a thickness of about 3 mm, which is around the ball-park value of $35 \text{ fs}^2/\text{mm}$ for Borosilicate and Fused silica glass. Switching back to the FSAC but making the beam go through the window, the results were consistent since it was found that $n = 10$ bounces gives the near-zero chirp trace (see figure 9) With this new trace, the pulse

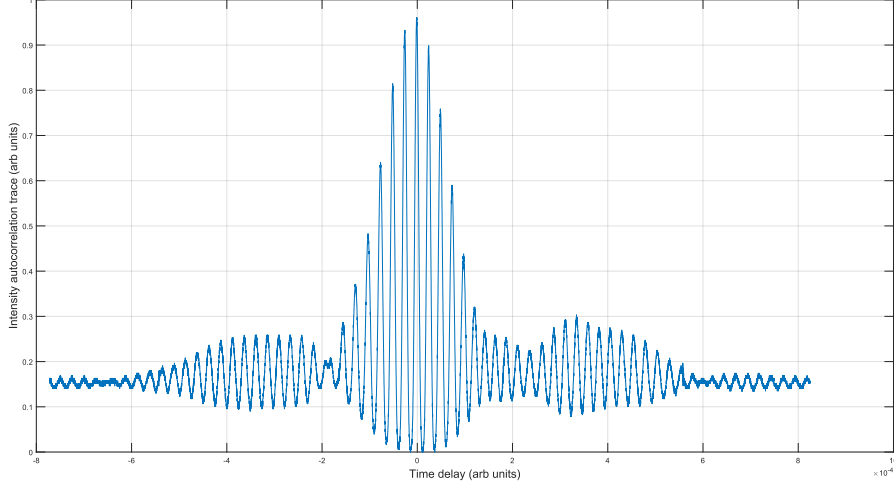


Figure 9: Interferometric trace for $n = 10$ with the beam crossing a glass window.

duration is

$$\text{Gaussian pulse FWHM : } 12.6613 \text{ fs, } \text{sech}^2 \text{ pulse FWHM : } 11.6045 \text{ fs} \quad (2)$$

5 Issues

There are at least two things wrong with the traces shown in figure 8:

Total GDD. If we add the GDD of each individual element at the central wavelength of 756 nm

$$\text{GDD}_T = \text{GVD}_{\text{air}} * z_d + 3\text{GDD}_m + \text{GDD}_{\text{FSAC}} + \text{GDD}_{\text{cm}} * n \quad (3)$$

where $\text{GDD}_{\text{air}} = 23.781 \text{ fs}^2/\text{m}$ according to the refractive index database, $z_d = 98 \text{ cm}$ is the distance from the laser pinhole to the FSAC pinhole, $\text{GDD}_m \simeq -1 \text{ fs}^2$ from Newport's web page, $\text{GDD}_{\text{FSAC}} = 230 \text{ fs}^2$ from the manual, $\text{GDD}_{\text{cm}} = -37.4 \text{ fs}$ according to the chirped mirrors raw data, and n is the number of bounces. Equation (3) seems to indicate, therefore, that the number of bounces required to get the nearest-to-zero chirp should be $n = 6$ rather than $n = 8$.

Trace wings. Both the FSAC's manual and web page show the interferometric trace that can be seen in figure 10. This trace, even though shows a little bit of symmetric wings on either side of the trace, they aren't far apart like the ones that always show up in my measurements, which are also asymmetric (see figure 8). From reference [1], if there is a second pulse close enough to the main pulse, this can cause the wings I observed in the trace of figure 8.

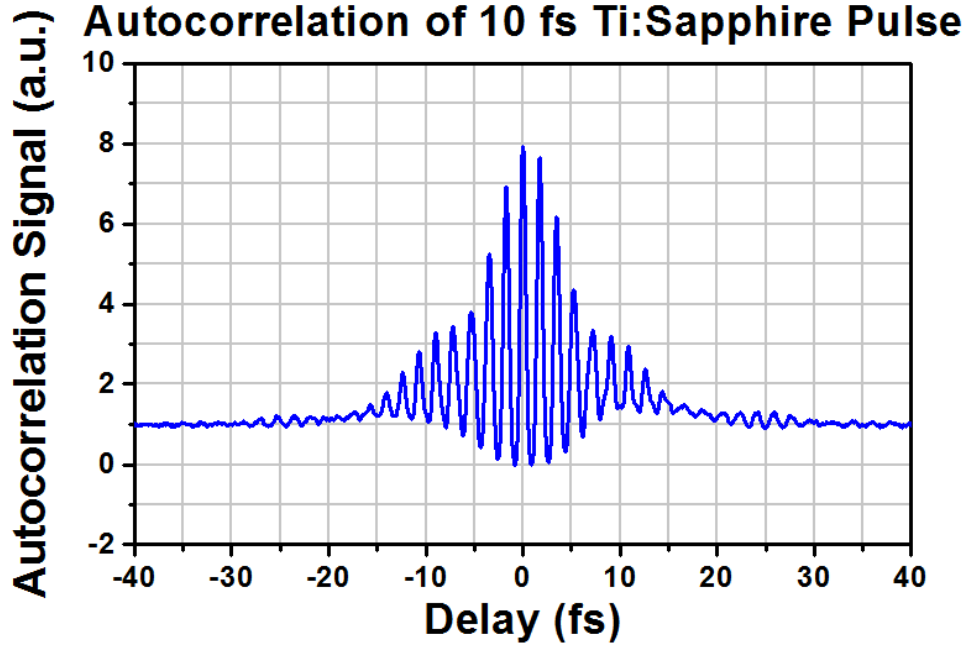


Figure 10: Autocorrelation trace provided by Thorlabs.

5.1 Code

Following Agrawal [2], a transform limited Gaussian pulse with duration T_0 and frequency ω_0 that travelled a distance z_d through a medium with GVD of value β_2

$$E(z_d, t) = \frac{E_0}{\sqrt{2\pi}T_1} e^{-\frac{t^2}{2T_1^2}} \cos\left(\omega_0 t + \frac{\omega_0 \beta_2 z_d}{T_0^3}\right), \quad (4)$$

where

$$\beta_2 = \left. \frac{\partial^2 k}{\partial \omega^2} \right|_{\omega=\omega_0}, \quad (5)$$

and the new pulse duration T_1 is given by

$$T_1 = \sqrt{1 + \left(\frac{\beta_2 z_d}{T_0^2}\right)^2} T_0. \quad (6)$$

We use these expressions to model a Gaussian pulse as it travels through a medium with $GDD = \beta_2 z_d$. The problem is that the pulse that we have in the lab is not Gaussian (see figure 11). Therefore, we fit the pulse with three Gaussian functions (see figure 12) and use the fit as the intensity profile to get the pulse electric field and therefore compute the nonlinear autocorrelation trace given by the expression

$$S_q(\tau) = \int_{-\infty}^{\infty} [E(t) + E(t + \tau)]^4 dt \quad (7)$$

to obtain the trace that can be seen in figure 13. Note that, even though the “shoulders” are present in the trace, they are way bigger than those that appear in the experimental trace (see figure 8). If, however, we tune the fitting a bit by moving the two lateral Gaussians of figure 12 away from one another and making them smaller in amplitude (see figure 14), we can get a trace that looks more similar (see figures 15) to the experimental trace. With this last image we can argue that the wings in the experimental interferometric trace may originate from the shoulders on the time pulse shape.

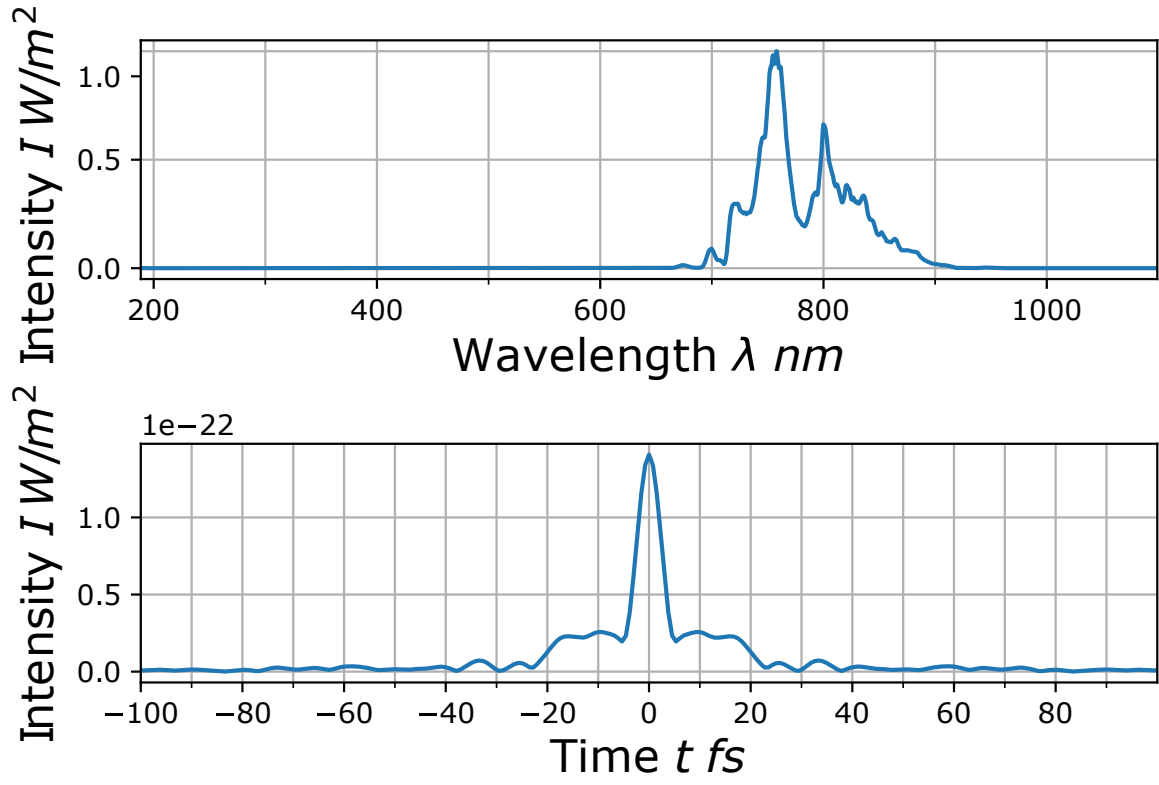


Figure 11: Top: Laser spectrum. Bottom: Inverse Fourier transform of the spectrum to get the time pulse shape.

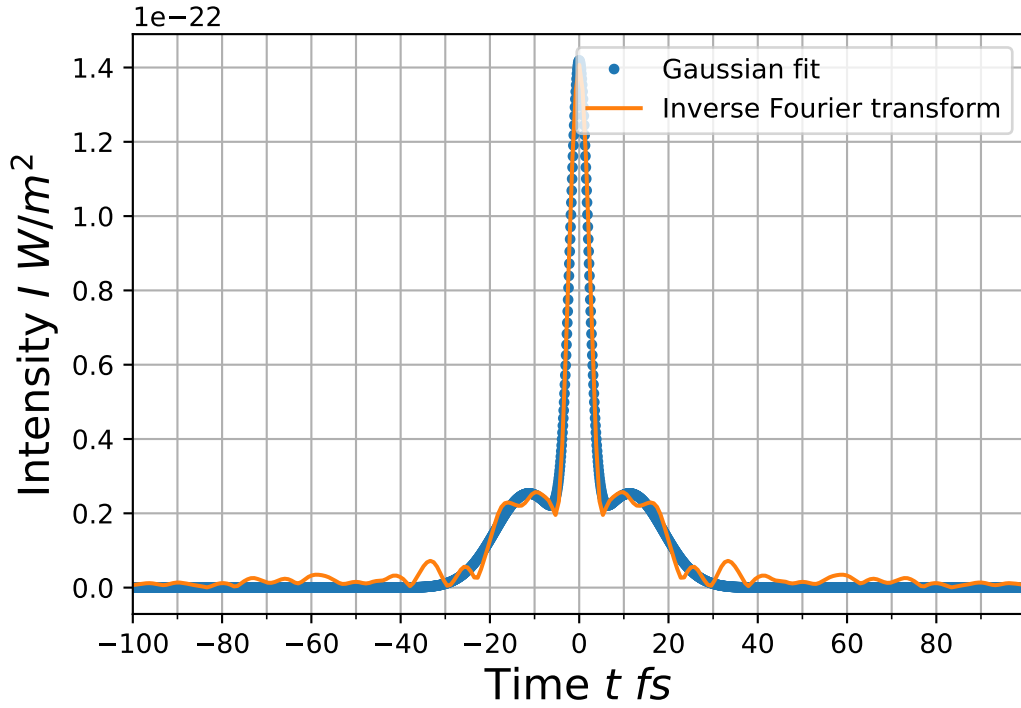


Figure 12: Fitting of pulse shape recovered from the inverse Fourier transform of the spectrum by using three Gaussian functions.

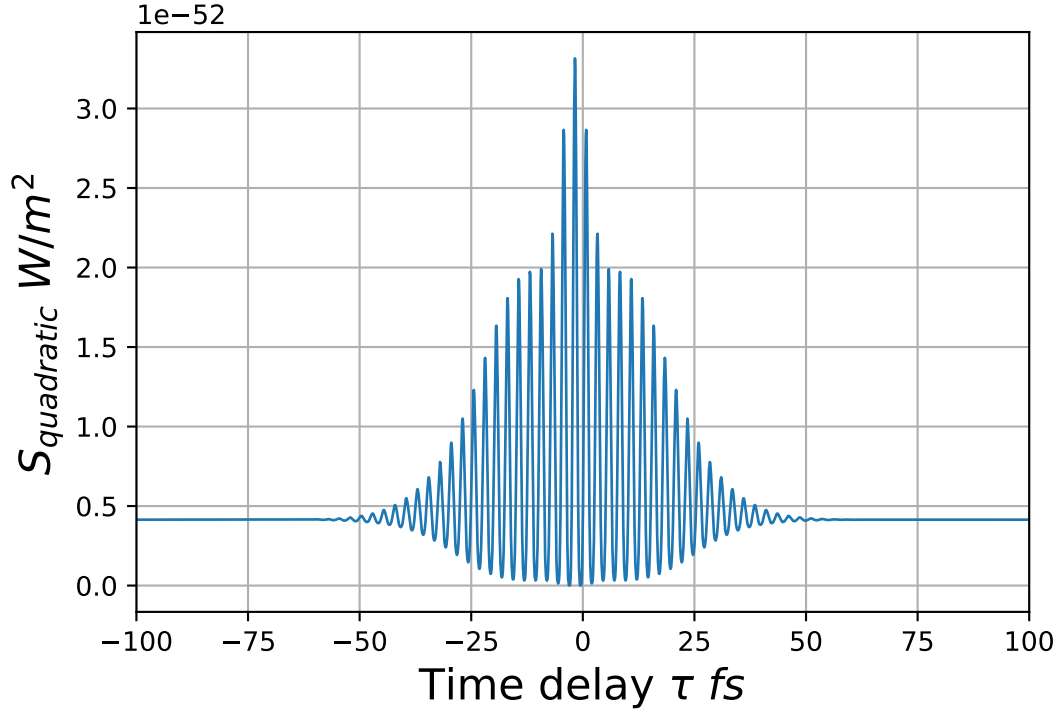


Figure 13: Non-linear detector trace S_q for a chirp-less three Gaussian fitted pulse.

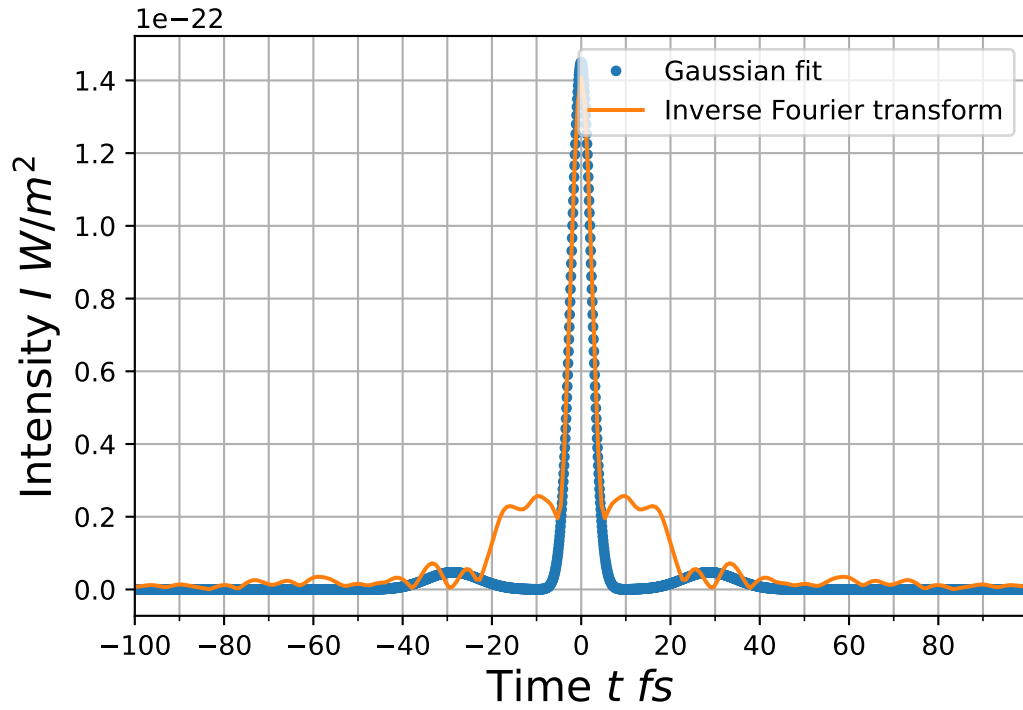


Figure 14: Interferometric trace obtained by adjusting the lateral

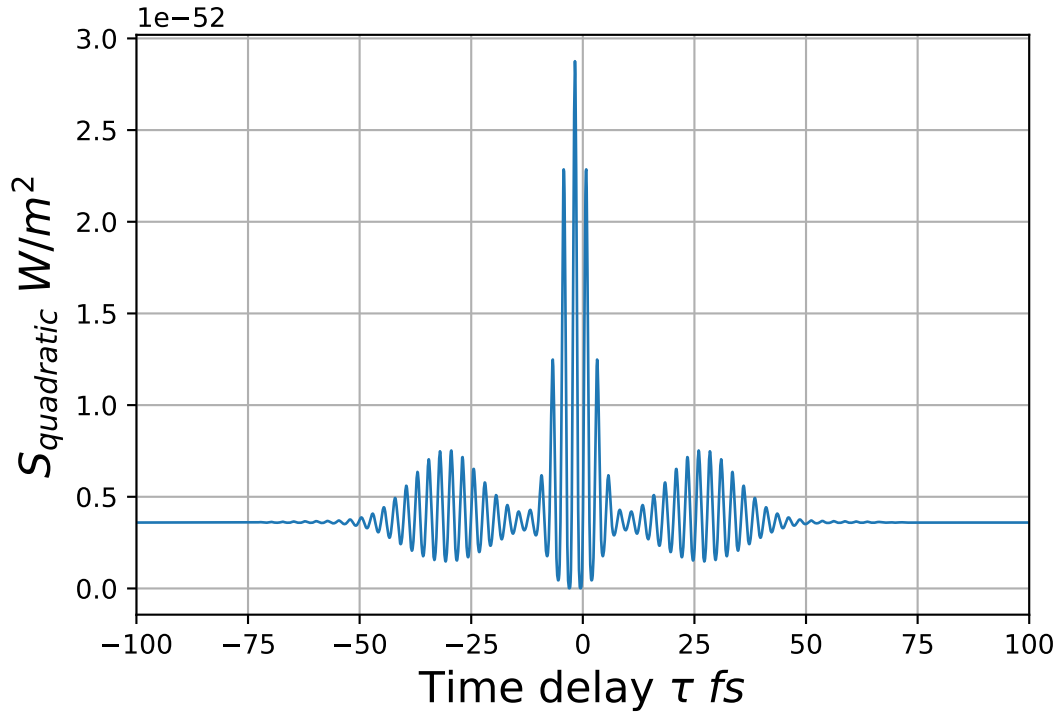


Figure 15: Interferometric trace obtained by adjusting the lateral

Even when this last figure looks promising, when I add chirp to it, the lateral wings disappear from the interferometric trace, and thus the origin of the wings remains uncertain, since they are always present regardless of the amount of chirp in the pulse (see figures 2-7).

References

- [1] Rick Trebino. *Frequency-Resolved Optical Gating: The Measurement of Ultrashort Laser Pulses*. Springer New York, NY, 2012. Figure 4.17, page 86.
- [2] Govind Agrawal. *Nonlinear Fiber Optics*. Elsevier Inc., 2013. Section 3.2.2 “Chirped Gaussian pulses”, page 62.